Category-Controlled Sketch Generation With Variational Autoencoders

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github.com/ashlabs/sketch-generation-vae

Overview

This project implements a category-controlled sketch generator for apples, mushrooms, and umbrellas using convolutional variational autoencoders (VAEs) trained from scratch in PyTorch. The goal is to generate recognizable variation from random latent vectors while making category selection explicit. A VAE was chosen because its encoder, probabilistic latent representation, and decoder support both reconstruction analysis and latent-space exploration. The reparameterization approach enables stochastic latent sampling while retaining a gradient path for optimization (Kingma & Welling, 2014).

Development began with an apple-only feasibility model, followed by a three-category conditional baseline and two architecture comparisons. The original 16-dimensional conditional model remained the leading candidate. Evaluation combined held-out losses, unfiltered samples, a separate category classifier, nearest-training-image comparisons, and latent interpolation. The result is an educational generative system with documented failures, rather than a production-quality illustration tool.

Dataset or Prompt Description

The source is Google Quick, Draw!, a public collection of drawings contributed by game participants. Its NumPy bitmap format provides centered 28 × 28 grayscale images (Google Creative Lab, n.d.). The downloaded apple, mushroom, and umbrella files contain 144,722, 142,167, and 124,084 images, respectively. These are human drawings, not AI-generated training data. Source URLs, file sizes, data types, and SHA-256 hashes are saved by the notebook.

Blank images and exact pixel duplicates are removed automatically before reproducible sampling. A shared hash inventory reserves images in apple, mushroom, and umbrella order, preventing exact duplicates across selected categories and partitions. The apple file contained no blank images and one exact duplicate. The original apple partition is preserved when adding

the other categories. Each category contributes 30,000 selected images, producing the balanced dataset in Table 1.

Table 1

Selected Dataset Partitions

Category	Training	Validation	Test
Apple	24,000	3,000	3,000
Mushroom	24,000	3,000	3,000
Umbrella	24,000	3,000	3,000
Total	72,000	9,000	9,000

The loader converts uint8 pixels to float32 intensities in [0, 1] and reshapes them to one-channel images. No random image augmentation is used. The selected bitmap files omit participant identifiers and recognition flags; exact deduplication therefore does not establish independence between artists or exclude near duplicates. The dataset supports simple object generation, but its limited resolution constrains fine stroke detail.

Model Design and Training Approach

Architecture and Objective

The encoder uses two 3 × 3 stride-2 convolutions with 32 and 64 channels, reducing the image to 7 × 7 feature maps. A 128-unit dense layer feeds separate posterior mean and log-variance heads. The conditional baseline concatenates a three-category one-hot label with the encoder features and with the sampled latent vector at the decoder input. The complete 16-dimensional conditional baseline contains 855,201 trainable parameters.

The decoder maps the latent vector and label through dense layers to 64 feature maps at 7 × 7, then applies nearest-neighbor upsampling and convolutions to restore 28 × 28 pixels. It returns logits; sigmoid intensities are used for display and generated-image evaluation.

Sampling combines the posterior mean with scaled normal noise, following the VAE reparameterization formulation (Kingma & Welling, 2014).

The objective is reconstruction binary cross-entropy (BCE) plus KL divergence to a standard normal prior, with KL weight 1. Reconstruction error is summed over 784 pixels, KL over latent dimensions, and both are averaged over images. The Gaussian KL regularizer follows the VAE framework (Kingma & Welling, 2014); grayscale BCE targets are used here as a soft-target pixel objective, not as a claim that observed intensities are binary. Lower loss does not necessarily mean better-looking sketches.

Controlled Experiments and Reproducibility

All VAEs use Adam with learning rate 0.001 and batch size 128. The apple feasibility model trains for five epochs and continues to 20 total, preserving optimizer state. Three fresh conditional models each train for 20 epochs. The capacity experiment changes latent dimensions from 16 to 32; the conditioning experiment supplies category maps before every decoder convolution while retaining 16 dimensions. Data splits, loss weights, convolution widths, and training budget remain fixed.

Validation uses fixed image order and a separate CPU noise generator reset to seed 12345. Checkpoints are selected by lowest aggregate validation loss; training and per-category histories are retained. Table 2 shows that neither change improved the observed validation objective. The comparisons use one training seed, and different architectures alter initialization and optimization. They do not prove that smaller latent spaces or initial-layer conditioning are generally superior.

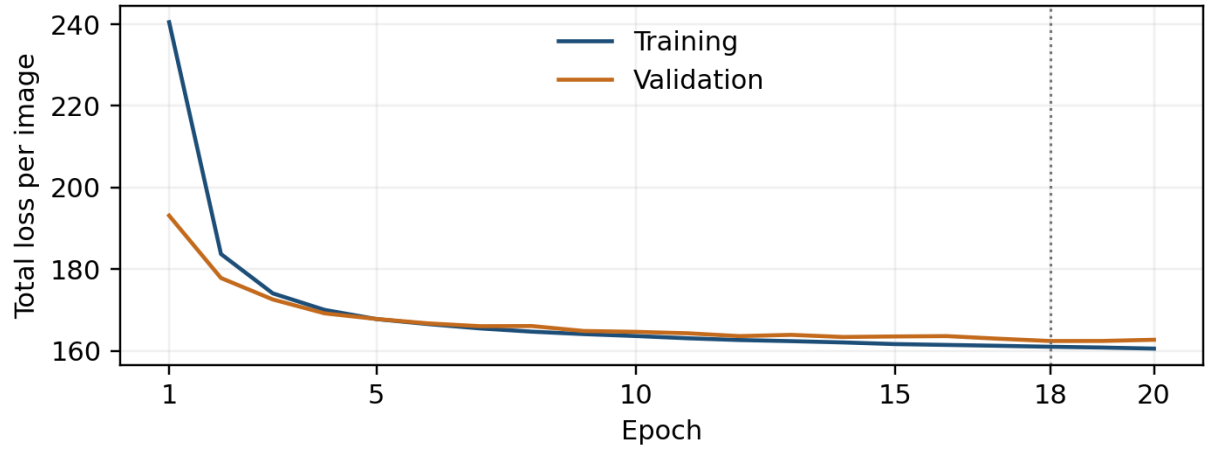
Table 2

Conditional Model Comparison at Each Best Validation Epoch

Model	Epoch	Total	Reconstruction	KL
Original, latent 16	18	162.4108	140.1169	22.2939
Original, latent 32	20	168.3166	149.1604	19.1561
Spatial, latent 16	20	164.2497	143.6237	20.6260

Figure 1

Training and Validation Total Loss for the Selected Architecture



Note. Curves cover the original 16-dimensional conditional model. Checkpoint selection uses validation total loss; the final epoch is not automatically selected.

The completed environment used Python 3.14.0 and PyTorch 2.15.0 on an Apple M5 Pro with 24 GiB unified memory and MPS acceleration. The working environment is recorded in requirements.txt. A fresh-kernel rerun executed all 44 code cells in order without saved errors and reproduced the reported metrics at displayed precision. New timestamped directories preserve prior runs; raw data and model checkpoints remain outside Git. Results may differ across devices and software versions.

Output Evaluation and Interpretation

Held-Out Reconstruction Generalization

The original conditional baseline was fixed at its epoch-18 minimum-validation-loss checkpoint before test evaluation. Its validation total was 162.4108 and its held-out total was 162.3114 across 9,000 images. Test reconstruction loss was 140.0546 and KL was 22.2567. The close validation and test values provide no evidence of a large partition gap under this objective; they do not establish perceptual quality or independent-artist generalization.

Table 3

Selected VAE Test Loss by Category

Category	Images	Total	Reconstruction	KL
Apple	3,000	155.1924	133.4294	21.7630
Mushroom	3,000	168.6041	145.5670	23.0371
Umbrella	3,000	163.1375	141.1674	21.9701

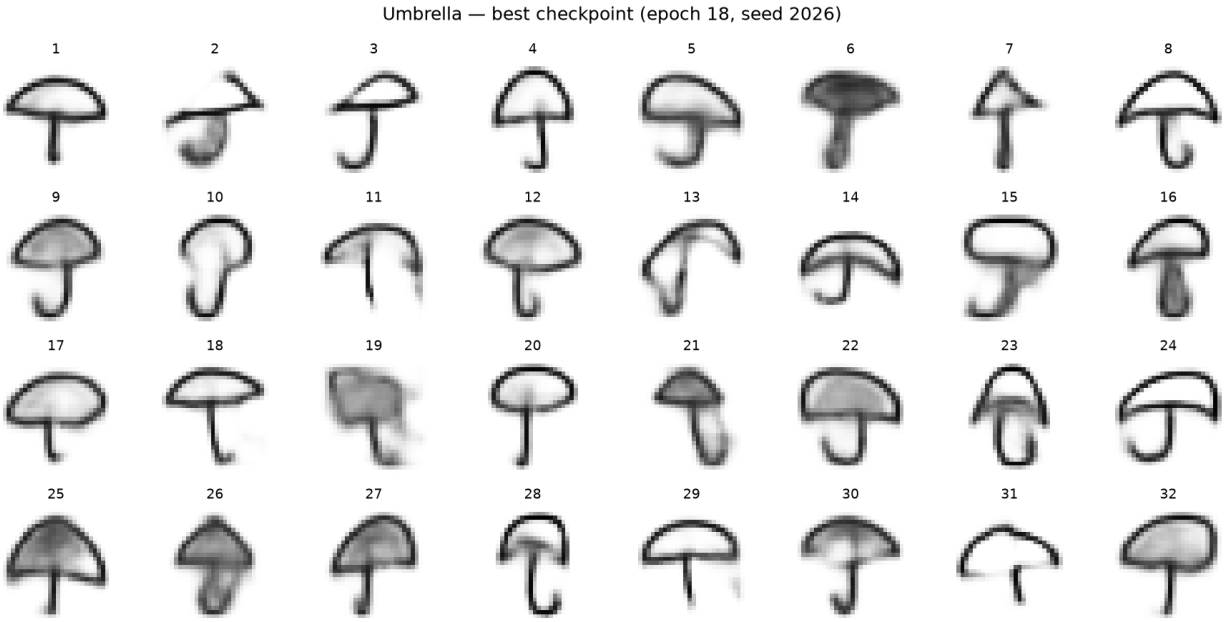
Apples have the lowest pixel-based objective. Cross-category differences can also reflect drawing complexity and stroke density, so they are not a direct ranking of human-perceived output quality. Fixed validation reconstructions preserve broad proportions and tilt, while thin stems, leaves, and angular details are softened.

Prior Samples and Category Recognition

Prior sampling evaluates new outputs rather than reconstructions of individual inputs. The selected checkpoint was reviewed using 32 unfiltered samples per category with seed 2026. Apple sample 13 has an incomplete outline; mushroom samples 11 and 13 have broken structures. Several umbrella samples have clear curved handles, but others resemble mushrooms. Figure 2 retains all umbrella review samples, including defects, instead of selecting only attractive examples.

Figure 2

Unfiltered Umbrella Samples From the Selected Epoch-18 Checkpoint



Note. The grid contains all 32 samples from review seed 2026. For example, samples 3 and 8 show curved handles, whereas samples 10 and 13 have ambiguous or malformed lower structures.

A separate 420,739-parameter CNN classifier was trained for five epochs on real training sketches only. It receives image pixels, not the requested label. Its lowest-validation-cross-entropy checkpoint was frozen before assessing generated images. Real-validation accuracy was 96.76% and real-test accuracy was 96.54%, providing context for its usefulness and imperfections as an evaluator.

Table 4

Frozen Classifier Evaluation

Category	Real-test recall	Generated agreement
Apple	98.33%	97.7%
Mushroom	96.57%	86.2%
Umbrella	94.73%	87.2%
Overall	96.54% accuracy	90.37% agreement

The generated evaluation uses 1,000 unfiltered images per category and fixed latent seed 2718. Of 289 disagreements, 266 are mushroom-umbrella swaps: 138 requested mushrooms are predicted as umbrellas and 128 requested umbrellas as mushrooms. This agrees with the visual ambiguity of their stems and handles. The 90.37% overall agreement measures classifier-based recognizability, not the fraction of visually good outputs. The classifier always selects one of three categories, cannot reject unknown content, and may behave differently on blurred generated images.

Similarity and Latent Variation

For the 96 fixed review samples, nearest-neighbor search compared each output with all 24,000 same-category training sketches using pixel MSE. Median distances were 0.03590 for apples, 0.04135 for mushrooms, and 0.04069 for umbrellas; none was zero. Displayed neighbors shared broad shapes but differed in stroke placement, proportions, and details. These observations do not rule out memorization: no matched validation-to-training reference distribution was measured, and small alignment changes affect pixel distance. Larger distances may reflect defects rather than useful novelty.

Linear interpolation between two fixed latent endpoints produced gradual changes in proportions, caps, stems, and handles while holding each label constant. Mushroom stems and

umbrella handles remained distinguishable along the inspected path, but apple leaf details became smeared toward an endpoint. One path illustrates local decoder behavior, not complete latent-space coverage. Unfiltered sample grids demonstrate some variation, but the study does not quantify overall distribution coverage or exclude mode loss.

Ethical Considerations and Responsible Use

Creative ownership is directly relevant because training data consist of contributed human drawings. The source documentation identifies the dataset license as CC BY 4.0 (Google Creative Lab, n.d.). This license requires appropriate credit, a license link, and identification of modifications when sharing adapted material (Creative Commons, n.d.). The repository therefore attributes the dataset separately from its MIT code license. Public availability is not treated as evidence that the drawings were authored by this project, and generated outputs should be labeled as machine-generated.

Representational bias is a project-specific limitation: voluntary game participation and prompted object categories do not establish representative coverage of cultures or artistic styles. The bitmap subset does not permit artist-level analysis, so the project cannot quantify that bias. A high category-agreement score should not be presented as cultural validity, originality, or universal human recognition.

Responsible presentation retains malformed examples, reports unsuccessful experiments, separates recognizability from visual quality, and avoids production-quality claims. The nearest-neighbor diagnostic is not used to certify creative originality. The narrow three-object scope reduces the range of potential misuse compared with unrestricted generation, but does not eliminate obligations to attribute source material or describe limitations honestly.

Limitations and Future Improvements

The main output limitations are blurred or smeared strokes, incomplete contours, and category ambiguity. The 28 × 28 representation cannot support details absent from the source images, and enlarging displayed pixels does not add information. The objective favors pixel reconstruction and latent regularization rather than directly optimizing human-rated sketch quality. The larger latent and repeated-conditioning experiments did not resolve these problems in their recorded training budgets.

Evidence is limited to single-seed training comparisons, one posterior sample per evaluation image, a classifier proxy, 96 nearest-neighbor queries, and one interpolation path. Exact deduplication does not control artist identity or near duplicates. Future work should first add repeated-seed comparisons, a prespecified human rating protocol, and reference distributions for similarity analysis. Higher-resolution or stroke-sequence modeling could then be tested in a separate development study with a new evaluation plan. The completed test results should not be reused to guide further tuning.

The project demonstrates a reproducible conditional generator and a transparent evaluation process. Its strongest result is recognizable category control on simple sketches, supported by held-out reconstruction behavior and explicit failure analysis. It remains an educational model whose visual fidelity and evaluation breadth leave room for improvement.

References

Creative Commons. (n.d.). *Attribution 4.0 International (CC BY 4.0).* https://creativecommons.org/licenses/by/4.0/

Google Creative Lab. (n.d.). *The Quick, Draw! dataset* [Data set]. GitHub. https://github.com/googlecreativelab/quickdraw-dataset

Kingma, D. P., & Welling, M. (2014). *Auto-encoding variational Bayes.* International Conference on Learning Representations. https://arxiv.org/abs/1312.6114